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CASE STUDY 2

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Reusable Digital Platform for Designing Battery Energy System

Problem Description

Designing a battery energy system requires engineers to understand how cells, electrical connections, cooling, sensing, protection, packaging, and controls work together. Improving one parameter, such as capacity or charging speed, may negatively affect temperature, safety, weight, battery life, or cost. Your challenge is to create an innovative digital platform that helps students and engineers virtually design, test, compare, and improve battery energy system configurations. The platform should make complex system interactions and design trade-offs easier to understand while supporting the development of safe, efficient, reliable, and affordable battery systems.

Underlying Challenge

- Cell chemistry, electrical, thermal, mechanical, safety and control data are distributed across disciplines.
- The design must balance energy, temperature, charging time, safety, size and cost.
- Cell variation, ageing and operating conditions propagate unpredictably from cell to pack and system levels.
- Limited availability of test data, software access and computing resources.
- A full battery pack is too large for a short hackathon, so the scope must remain manageable.

Solution Goal

- Develop an integrated digital platform to virtually configure, visualize, analyze, and compare battery energy system designs from cell to system level.
- Integrate electrical, thermal, mechanical, sensing, protection, control, and energy-management considerations within a unified engineering workflow.
- Enable consistent exchange of parameters and results between subsystem models, allowing design changes and their system-level impacts to be evaluated.
- Identify interface conflicts, incompatible assumptions, safety-limit violations, and performance trade-offs across battery subsystems.
- Maintain traceability between requirements, component selections, assumptions, models, results, and validation evidence while recommending a balanced design with clearly stated risks and limitations.

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Key Points the Solution should address

- Select a manageable scope, such as a small cell group, module, cooling arrangement or monitoring function.
- Define the application, basic requirements and expected operating conditions.
- Consider cells, electrical connections, temperature control, sensing, protection and basic controls.
- Use available datasheets, simple calculations, simulations or safe low-voltage experiments.
- Compare at least two design options against a basic starting design.
- Measure a few relevant indicators such as usable energy, temperature, charging time, mass, size or estimated cost.
- Document assumptions, data sources, safety precautions, results and limitations.
- Present a simple prototype or visual demonstration and explain the recommended next step.

Evaluation Criteria

Criterion	Weight	Assessed on
Problem understanding	15%	Clear application, scope, requirements & explanation of the battery challenge.
Desirability	15%	Value of the idea to a user and relevance of the problem being solved.
Technical feasibility	20%	Simple, safe and achievable design using available time, data and resources.
Engineering quality	15%	Sound calculations, assumptions and understanding of system interactions.
Prototype & evidence	15%	Working demonstration or model with measurable and repeatable results.
Innovation	10%	Creative but understandable improvement over the starting design.
Business viability	10%	Potential value in cost, performance, safety, usability or scalability.